

Notations

1. Greek Alphabets

Uppercase	Lowercase	Name
A	α	Alpha
B	β	Beta
Γ	Γ	Gamma
Δ	Δ	Delta
E	ϵ	Epsilon
Z	ζ	Zeta
H	η	Eta
Θ	θ	Theta
I	ι	Iota
K	κ	Kappa
Λ	λ	Lambda
M	μ	Mu
N	ν	Nu
Ξ	ξ	Xi
O	$\omicron$	Omicron
Π	π	Pi
P	ρ	Rho
Σ	σ	Sigma
T	τ	Tau
Υ	υ	Upsilon
Φ	φ	Phi
X	χ	Chi
Ψ	ψ	Psi
Ω	ω	Omega

2. Logical Notation

Symbol	Name	Read As	Example
$\neg$	Negation	Not	$\neg P$ (Not P)
$\wedge$	Conjunction	And	$P \wedge Q$ (P and Q)
$\vee$	Disjunction	Or	$P \vee Q$ (P or Q)
$\oplus$	Exclusive Or	Either...or	$P \oplus Q$ (Either P or Q)
$\rightarrow$ or $\Rightarrow$	Implication	If...then/Implies	$P \rightarrow Q$ (If P then Q)
$\leftrightarrow$ or $\Leftrightarrow$	Biconditional	If and only if	$P \leftrightarrow Q$ (P if and only if Q)
$\forall$	Universal Quantifier	For Every	$\forall x P(x)$
$\exists$	Existential Quantifier	There exists/ For some	$\exists x P(x)$
$\nexists$	Non Existential Quantifier	There exists/ For some	$\nexists x P(x)$
$\exists!$	Uniqueness Quantifier	There exists unique	$\exists! x P(x)$
$\top$	Tautology	Always true	
$\perp$	Contradiction	Always false	
$\vdash$	Syntactic Turnstile	Proves/Yields (Provable via inference rules)	
$\nvdash$	Non provability	Does not prove	
$\models$	Syntactic Double Turnstile	Entails/Is a model of (True in all models)	
$\not\models$	Non entailment	Does not entail	
$\equiv$	Logical Equivalence	Is logically equivalent to	
$\in$	Element of	Belongs to	$x \in A$ (x belongs to A)

$\notin$	Not Element of	Is not an element of	$x \notin A$ (x is not an element of A)
$\subseteq$	Subset	Is a subset of	$P \subseteq Q$ (P is a subset of Q)
$\subset$	Strict Subset	Is a proper subset of	$P \subset Q$ (P is a proper subset of Q)
$\not\subset$	Not a Subset	Is not a subset of	$P \not\subset Q$ (P is not a subset of Q)
$\emptyset$	Empty Set	Is an empty set	$P = \emptyset$ (P is an empty set)
$\cup$	Union	union	
$\cap$	Intersection	intersection	
$\setminus$	Set Difference	minus	